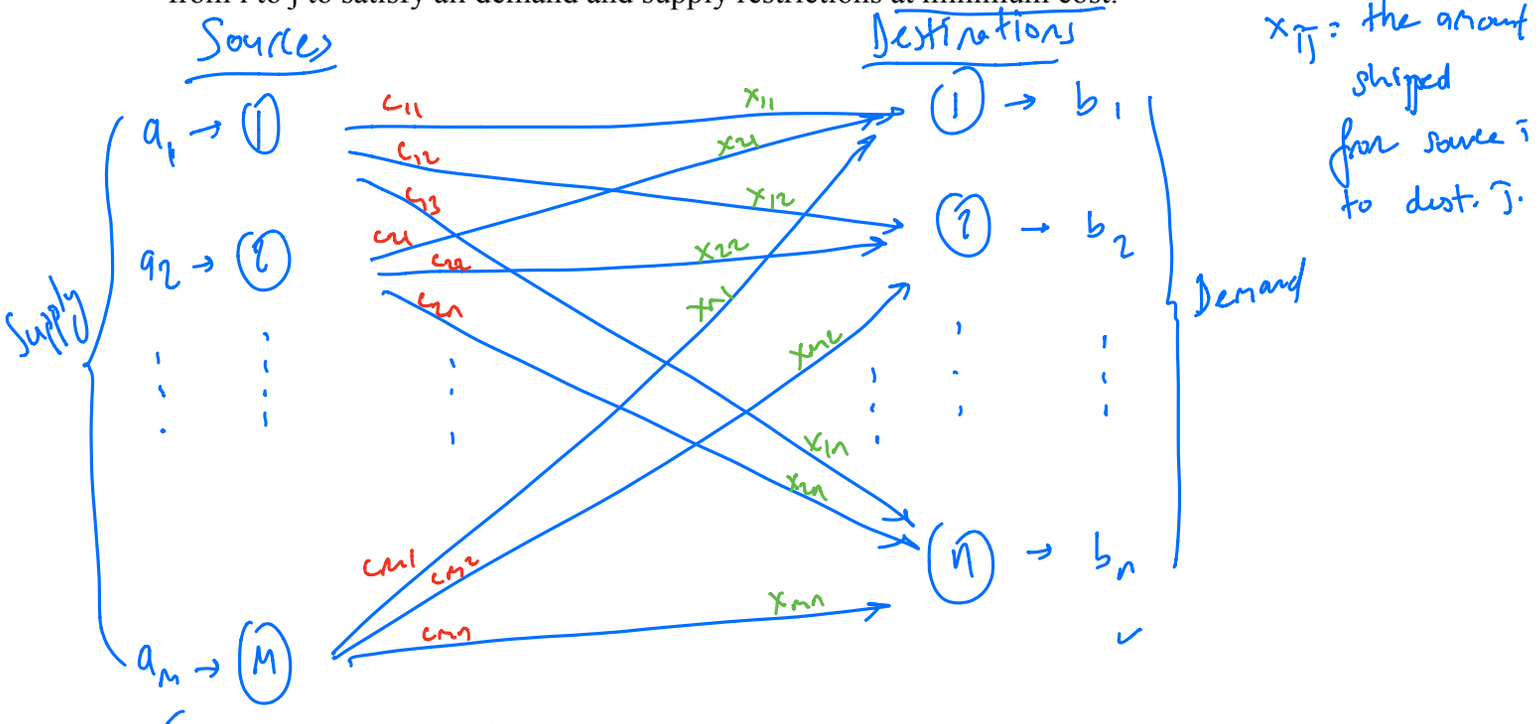
TRANSPORTATION AND ASSIGNMENT PROBLEMS

Definition of the Transportation Problem

There are m sources with supply a_i (i=1,…, m) and n destinations with demand b_j (j=1,…,n) and the unit cost of shipping from i to j is c_{ij}. What are the optimum quantities to be transported from i to j to satisfy all demand and supply restrictions at minimum cost.



$$\text{minimize} \quad \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} \cdot x_{ij}$$

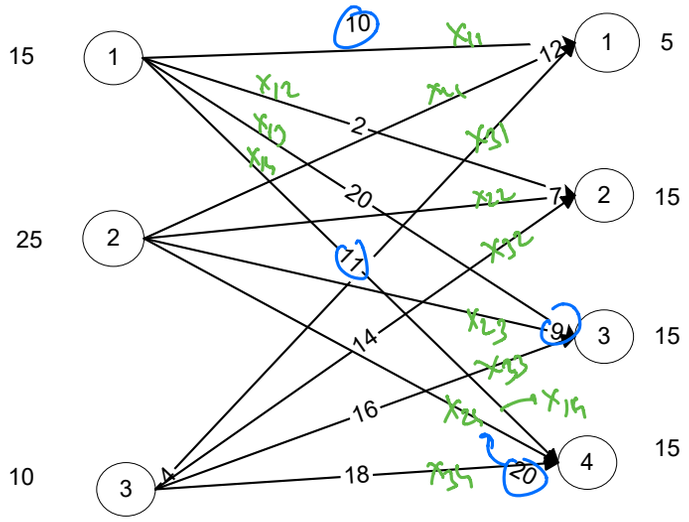
$$\text{s.t} \quad \sum_{j=1}^{n} x_{ij} = a_i \qquad \forall i \quad i = \{1, 2, \dots, m\}$$

$$\sum_{i=1}^{m} x_{ij} = b_j \qquad \forall j \quad j = \{1, 2, \dots, n\}$$

$$x_{ij} \geq 0$$

Example

Sunray ships truckloads of grain from 3 silos to 4 mills. The supply (in truckloads) at silos, the demand at mills and the unit transportation cost from silos to mills are shown below.



Let x_{ij} = no. of truckloads shipped from silo i to mill j.

The corresponding LP is as follows:

minimize $10x_{11} + 2x_{12} + 20x_{13} + 11x_{14} + 12x_{21} + 7x_{22} + 9x_{23} + 20x_{24} + 4x_{31} + 14x_{32} + 16x_{33} + 18x_{34}$

s.t $x_{11} + x_{12} + x_{13} + x_{14} = 15$

$x_{21} + x_{22} + x_{23} + x_{24} = 25$

$x_{31} + x_{32} + x_{33} + x_{34} = 10$

$x_{11} + x_{21} + x_{31} = 5$

$x_{12} + x_{22} + x_{32} = 15$

$x_{13} + x_{23} + x_{33} = 15$

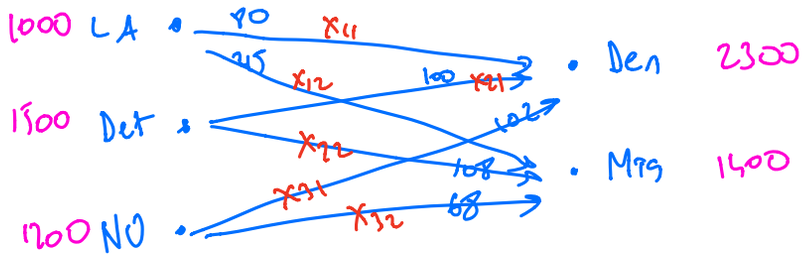
$x_{14} + x_{24} + x_{34} = 15$

$x_{ij} \geq 0$

Example:

MG Auto has three plants in Los Angeles, Detroit, and New Orleans and two major distribution centers in Denver and Miami. The quarterly capacities of the three plants are 1000, 1500, and 1200 cars, and the demands at the two distribution centers for the same period are 2300 and 1400 cars. The mileage chart between the plants and the distribution centers are given in the following table:

Mileage Chart	Denver	Miami
Los Angeles	1000	2690
Detroit	1250	1350
New Orleans	1275	850



The trucking company in charge of transporting the cars charges 8 cents per mile per car. Thus the transportation costs per car are on the different routes, rounded to the closest dollar are computed as follows:

Transportation Cost per Car	Denver	Miami
Los Angeles	$80	$215
Detroit	$100	$108
New Orleans	$102	$68

The LP model of the problem is

$$\text{Minimize } z = 80x_{11} + 215x_{12} + 100x_{21} + 108x_{22} + 102x_{31} + 68x_{32}$$

$$x_{11} + x_{12} = 1000$$

$$x_{21} + x_{22} = 1500$$

$$x_{31} + x_{32} = 1200$$

$$x_{11} + x_{21} + x_{31} = 2300$$

$$x_{12} + x_{22} + x_{32} = 1400$$

The special structure of the transportation problem allows a compact representation of the problem using the **transportation tableau** format as given in the next table:

	Denver	Miami	Supply
Los Angeles	80 $x_{11} = 1000$	215 $x_{12} = 0$	1000
Detroit	100 $x_{21} = 1300$	108 $x_{22} = 200$	1500
New Orleans	102 $x_{31} = 0$	68 $x_{32} = 1200$	1200
Demand	2300	1400	

Optimum solution: $x_{11} = 1000$, $x_{21} = 1300$, $x_{22} = 200$, $x_{32} = 1200$

This model can be solved by simplex method. However, the special structure of the problem (all constraint coefficients are 0 or 1), allows us to develop and use a special (and simpler) algorithm: The Transportation Simplex. However, we are going to skip this algorithm.

Unbalanced Transportation Problem

In the previous example total supply = total demand. It was a balanced problem. If this is not the case, we add a dummy source or destination (which one is less) and assign zero shipping costs to the cells between the dummy node and the others. Then supply or demand amount of the dummy node should be equal to the difference (this way, the network will be balanced). Then, we can use the regular transportation algorithm.

In the MG model, suppose that the Detroit plant capacity is 1300 cars (instead of 1500). Then total supply (=3500 cars) is less than the total demand (=3700 cars). Because the demand exceeds the supply, a dummy plant (source) with a capacity of 200 cars is added to balance the model.

	Denver	Miami	Supply
Los Angeles	80 1000	215 0	1000
Detroit	100 1700	108 0	1500 1300
New Orleans	102 0	68 1200	1200
Dummy Plant	0 0	200	200
Demand	2700	1400	

Excess Demand Example

	Den.	Mia	Dummy	Supply
LA	80 1000	215 0	0 0	1000
Det	100 900	108 200	0 400	1500
NO	102 0	68 1200	0 0	1200
Demand	2300 1900	1400	400	

Excess Supply Example

Next → Applications of the transportation problem
Next → Linear Assignment problem.

Applications of the Transportation Problem

Production – Inventory Control

Boralis is a backpack manufacturer. Sales occur from March to June with monthly demands 100, 200, 180 and 300. Production is also from March to June. It uses part time workers and monthly capacities differ: 50, 180, 280 and 270. A demand can be satisfied in three ways:

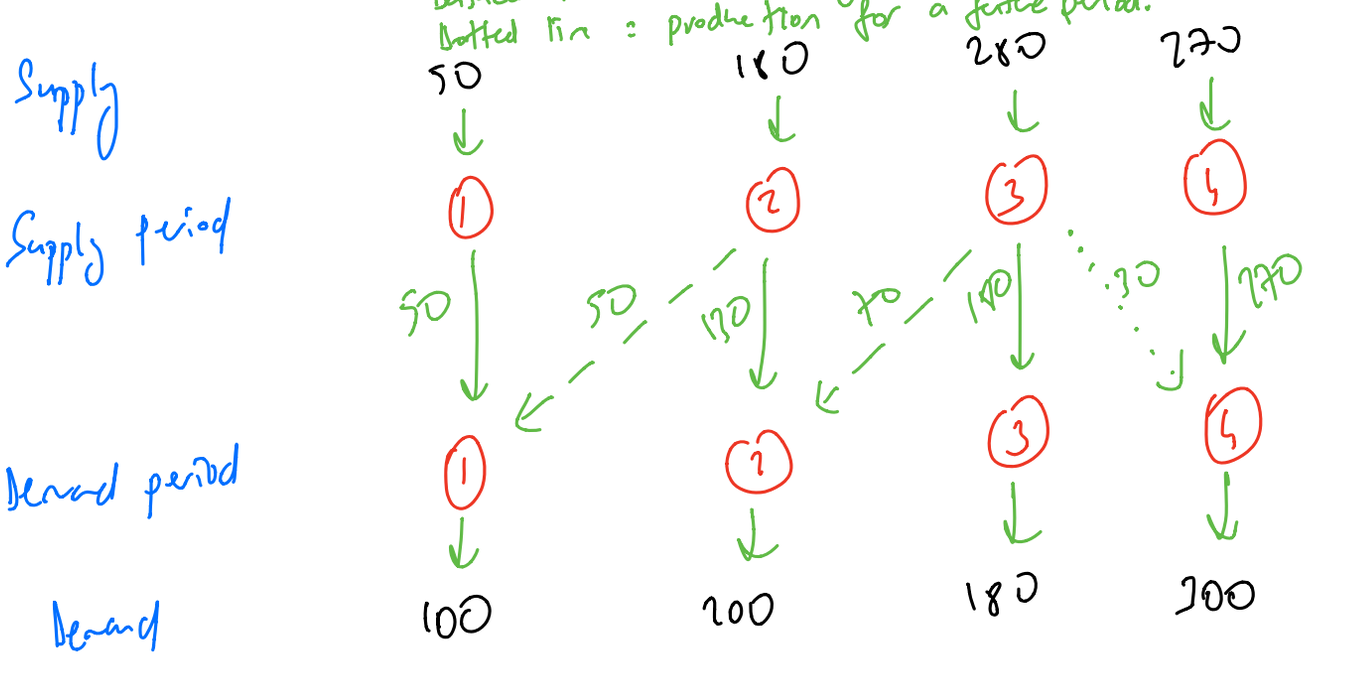
1) Current month's production (cost = \$40 / unit)
2) Surplus production in earlier months (cost = \$40 / unit + \$0.50 / unit*month)
3) Surplus production in later months (cost = \$40 / unit + \$2 / unit*month)

Formulate the problem as a transportation model.
Let c_{ij} = cost of producing in month i and selling in month j.

$$c_{ij}\begin{cases}40 & (i=j)\\ 40+0.5*(j-i) & (i<j)\\ 40+2*(i-j) & (i>j)\end{cases}$$

	March	April	May	June
Supply	50	180	280	270
Demand	100	200	180	300

Production \ Demand	March	April	May	June	Supply
March	50 (40)	0 (40.5)	0 (41)	0 (41.5)	50
April	50 (42)	130 (40)	0 (40.5)	0 (41)	180
May	0 (44)	70 (42)	180 (40)	30 (40.5)	280
June	0 (46)	0 (44)	0 (42)	270 (40)	270
Demand	100	200	180	300	



(Linear) Assignment Problem

It is a special case of the transportation problem where all supply and demand amounts are one.
"The best person for the job" is the famous example of the assignment problem.

Workers \ Jobs	1	2	.....	n
1	c11	c12	.....	c1n
2	c21	c22	.....	c2n
:	:	:		:
n	cn1	cn2	.....	cnn

c_{ij}: is the cost of worker i doing job j.

$$x_{ij} = \begin{cases} 1 & \text{if person } i \text{ is assigned to job } j \\ 0 & \text{otherwise} \end{cases}$$

The model is:

$$\min. \quad \sum_{i=1}^{n} \sum_{j=1}^{n} c_{ij} x_{ij}$$

$$\text{s.t} \quad \sum_{j=1}^{n} x_{ij} = 1 \qquad \forall i,\ i = \{1, 2, \dots, n\}$$

$$\sum_{i=1}^{n} x_{ij} = 1 \qquad \forall j,\ j = \{1, 2, \dots, n\}$$

$$x_{ij} = \{0, 1\}$$

Hungarian Algorithm:

Step 1: Subtract row minimum from row elements.
Step 2: Subtract column minimum from column elements.
Step 3: Identify optimum solution associated with zero elements.
If not possible, cover all zero elements with minimum number of lines. Subtract from uncovered elements the min of them and add that element to line intersections. Then, identify the optimum solution.

Example 5.4-1
Mr. Klyne asks his three children (John, Karen, Terri) how much they would like to be paid for the three jobs (mow, paint, wash). What is the optimum cost job assignment?

	Mow	Paint	Wash
John	15	10	9
Karen	9	15	10
Terri	10	12	8

→ Subtract row mins →

	Mow	Paint	Wash
John	6	1	0
Karen	0	6	1
Terri	2	4	0

Subtract column mins:

	M	P	W
J	6	0	0
K	0	5	1
T	2	3	0

Optimum assignment:

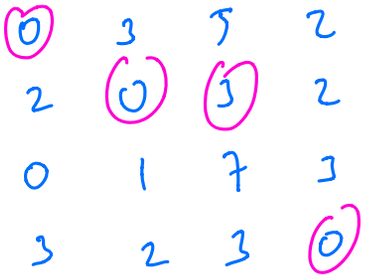
John → paint 10
Karen → mow 9
Terri → wash 8 cost = 27

Example

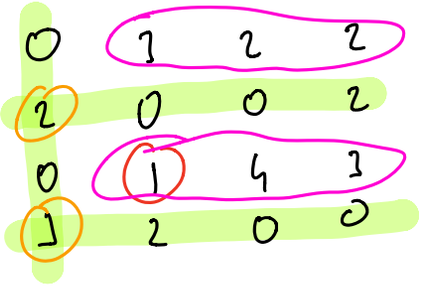
The previous example is extended to 4 children and 4 jobs, with the modified cost matrix as follows:

Child \ Job	1	2	3	4
1	1	4	6	3
2	9	7	10	9
3	4	5	11	7
4	8	7	8	5

→ minus row mins →

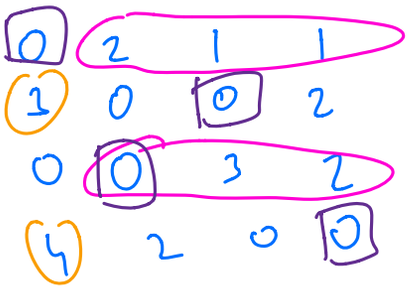


Minus column mins:



3 lines

The zeros are covered with (min) 3 lines as shown. The min of the uncovered entries is 1. Subtract 1 from uncovered entries and add 1 to line intersections:

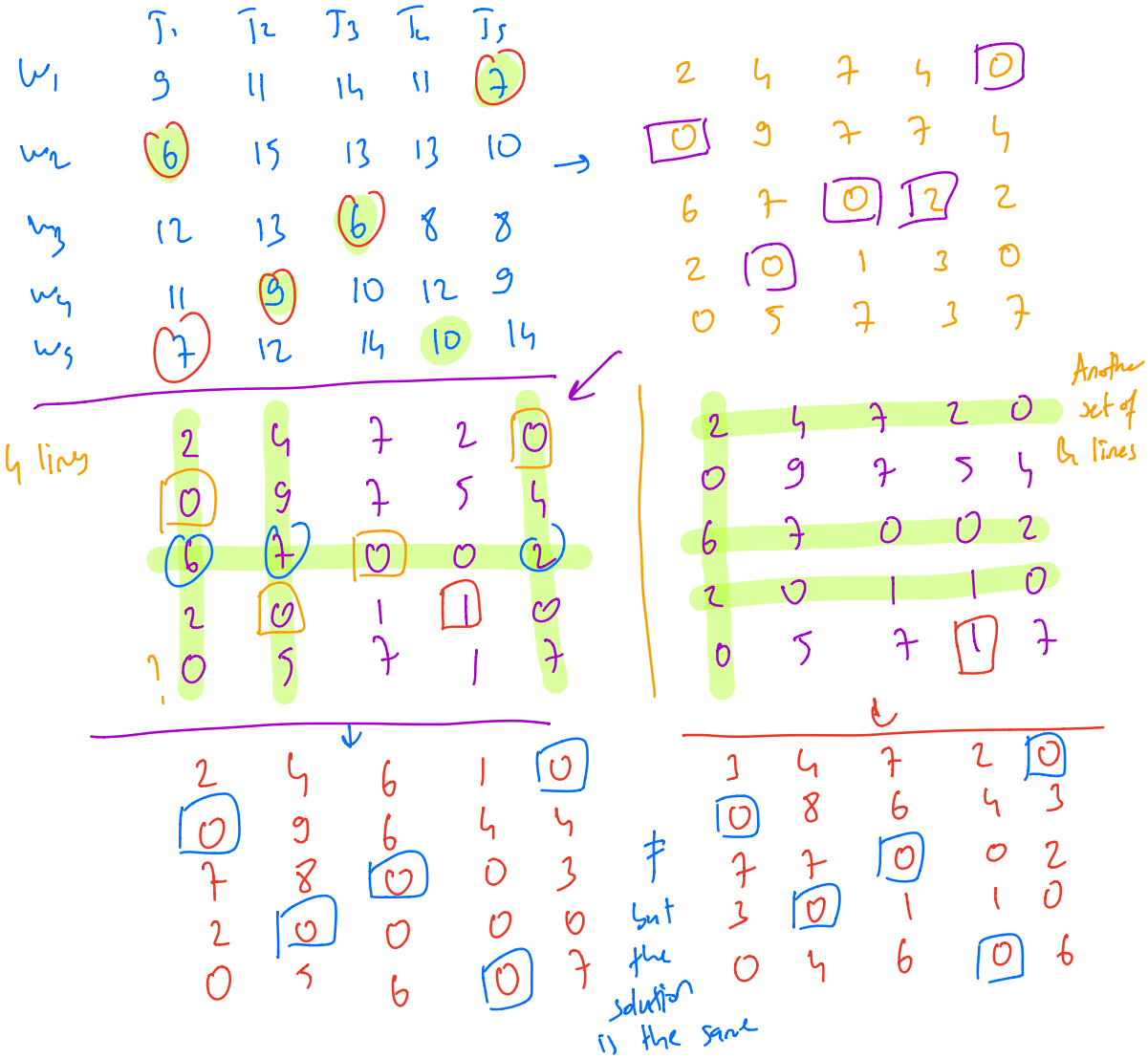


Optimal assignment;

$x_{11} = x_{23} = x_{32} = x_{44} = 1$ (the rest = 0)

cost = 1 + 10 + 5 + 5 = 21

- If a person cannot be assigned to a job, assign a large cost (M) to that cell.


J1 J2 J3 J4 J5
W1 9 11 14 11 7
W2 6 15 13 13 10
W3 12 13 6 8 8
W4 11 9 10 12 9
W5 7 12 14 10 14
4 lines
Another set of 4 lines
≠ but the solution is the same